

1.

Create a report that displays EMPNO,ENAME,DEPTNO of employees who work with that Employee.(Nested Sub Query)

```
SQL> select employee_id, last_name, department_id
2   from employees
3   where department_id =
4   (
5     select department_id
6     from employees
7     where employee_id = 100
8   )
9   and employee_id <> 100;
```

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID
101	Yang	90
102	Garcia	90

2.

Display the employees who earn more than the avg salary of EMP table.(Nested Sub Query).

```
SQL> select employee_id,last_name,salary
2   from employees
3   where salary>
4   (
5     select avg(salary)
6     from employees
7   );
```

EMPLOYEE_ID	LAST_NAME	SALARY
100	King	24000
101	Yang	17000
102	Garcia	17000
103	James	9000
108	Gruenberg	12008
109	Faviet	9000
110	Chen	8200
111	Sclarra	7700
112	Urman	7800
113	Popp	6900
114	Li	11000
120	Weiss	8000
121	Fripp	8200
122	Kaufling	7900
123	Vollman	6500
145	Singh	14000
146	Partners	13500
147	Errazuriz	12000
148	Cambraut	11000
149	Zlotkey	10500
150	Tucker	10000
151	Bernstein	9500
152	Hall	9000
153	Olsen	8000
154	Cambraut	7500
155	Tuvault	7000
156	King	10000
157	Sully	9500
158	McEwen	9000
159	Smith	8000
160	Doran	7500
161	Sewall	7000
162	Vishney	10500
163	Greene	9500
164	Marvins	7200
165	Lee	6000
168	Ozer	11500

168	Ozer	11500
169	Bloom	10000
170	Fox	9600
171	Smith	7400
172	Bates	7300
174	Abel	11000
175	Hutton	8800
176	Taylor	8600
177	Livingston	8400
178	Grant	7000
201	Martinez	13000
203	Jacobs	6500
204	Brown	10000
205	Higgins	12008
206	Gietz	8300

51 rows selected.

Display the ENAME,JOB who are Managers(Use Exists Operator)

3.

```
SQL> select e.last_name,e.job_id
2   from employees e
3  where exists
4  (
5    select 1
6    from employees m
7   where m.manager_id=e.employee_id);
```

LAST_NAME	JOB_ID
King	AD_PRES
Yang	AD_VP
Garcia	AD_VP
James	IT_PROG
Gruenberg	FI_MGR
Li	PU_MAN
Weiss	ST_MAN
Fripp	ST_MAN
Kaufling	ST_MAN
Vollman	ST_MAN
Mourgos	ST_MAN
Singh	SA_MAN
Partners	SA_MAN
Errazuriz	SA_MAN
Cambrault	SA_MAN
Zlotkey	SA_MAN
Martinez	MK_MAN
Higgins	AC_MGR

18 rows selected.

4.

Display the employees who earn less than the least salary of DEPTNO 10.(Use ALL operator).

```
SQL> select employee_id,last_name,salary
2   from employees
3  where salary<all
4  (
5    select salary
6    from employees
7   where department_id=10);
```

EMPLOYEE_ID	LAST_NAME	SALARY
185	Williams	4800
186	Jackson	4800
124	Mourgos	5800
184	Miller	6000
282	Davis	6000
173	Neer	6100
167	Banda	6200
179	Johnson	6200
166	Abel	6400
123	Vollman	6500
283	Jacobs	6500
165	Lee	6800
113	Popp	6900
155	Tuvault	7000
161	Seall	7000
178	Grant	7000
164	Marvin	7200
172	Bates	7300
171	Smith	7400
154	Cambrault	7500
160	Doran	7500
111	Sciarra	7700
112	Urman	7800
122	Kaufling	7900
120	Weiss	8000
153	Olsen	8000
159	Smith	8000
118	Chen	8200
121	Fripp	8300
286	Gietz	8300
177	Livingston	8400
176	Taylor	8400
175	Hutton	8800
183	James	9000
189	Faviet	9000
152	Hall	9000
158	McEwen	9000
157	Sully	9500
163	Greena	9500
151	Bernstein	9500
170	Fox	9600
284	Brown	10000
169	Bloom	10000

5.

Display the employees who have the same DEPTNO and MGR of a given employee, excluding that employee.(Use Nested Sub Query)

```
SQL> select employee_id, last_name
2 from employees
3 where (department_id, manager_id) =
4 (
5     select department_id, manager_id
6     from employees
7     where employee_id = 101
8 )
9 and employee_id <> 101;

EMPLOYEE_ID LAST_NAME
-----
102 Garcia
```

6.

Write a query that displays the employee number and name of all employees who work in a department with any employee whose name contains a R

```
SQL> select employee_id, last_name
2 from employees
3 where department_id in
4 (
5     select department_id
6     from employees
7     where last_name like '%R%'
8 );

EMPLOYEE_ID LAST_NAME
-----
120 Weiss
121 Fripp
122 Haurling
123 Vollman
124 Mourgos
125 Nayer
126 Mäkeläinen
127 Landry
128 Narverle
129 Bissot
130 Atkinson

EMPLOYEE_ID LAST_NAME
-----
131 Marlow
132 Olson
133 Hallin
134 Rogers
135 Gee
136 Philtanker
137 Ludwig
138 Stiles
139 Seo
140 Patel
141 Rajs

EMPLOYEE_ID LAST_NAME
-----
142 Davies
143 Ramos
144 Vargas
145 Taylor
146 Fleuss
147 Sullivan
148 Gensl
149 Sarchand
150 Bull
151 Dellinger
152 Cabrio

EMPLOYEE_ID LAST_NAME
-----
153 Chung
154 Dilly
155 Venzl
156 Perkins
157 Bell
158 Everett
159 McLeod
160 Jones
161 Walsh
162 Feeney
163 OConnell

EMPLOYEE_ID LAST_NAME
-----
164 Grant

45 rows selected.
```

The HR department needs a report that displays the ename, deptno, job of all employees whose work in NEW YORK.(Nest Sub Query)

7.

```

SQL> select last_name, department_id, job_id
2  from employees
3  where department_id in(
4  select department_id
5  from departments
6  where location_id=
7  (
8  select location_id
9  from locations
10 where city='New York'
11 )
12 );

no rows selected

```

8.

Modify the query so that the user is prompted for a LOC

```

SQL> select last_name, department_id, job_id
2  from employees
3  where department_id in
4  (
5      select department_id
6      from departments
7      where location_id =
8      (
9          select location_id
10         from locations
11         where city='&city'
12     )
13 );
Enter value for city: Bombay
old 11:      where city='&city'
new 11:      where city='Bombay'

no rows selected

```

9.

Create a report for HR that displays the name and salary of every employee who reports to King.(Nest Sub Query)

```

SQL> select last_name, salary
2  from employees
3  where manager_id in
4  (
5      select employee_id
6      from employees
7      where last_name = 'King'
8  );

```

LAST_NAME	SALARY
Yang	17000
Garcia	17000
Li	11000
Weiss	8000
Fripp	8200
Kaufling	7900
Vollman	6500
Mourgos	5800
Singh	14000
Partners	13500
Errazuriz	12000

LAST_NAME	SALARY
Cambrault	11000
Zlotkey	10500
Martinez	13000

14 rows selected.

10.

Write a query to display all the employees working with JAMES

```
2 from employees
3 where department_id in
4 (
5 select department_id
6 from employees
7 where last_name='James'
8 )
9 and last_name<>'James';
```

EMPLOYEE_ID LAST_NAME

```
-----
104 Miller
105 Williams
106 Jackson
107 Nguyen
```

11.

Display all the employees who earn less than the average salaries of their respective departments.(Correlated Sub Queries)

```
SQL> select employee_id,last_name,salary
2 from employees e
3 where salary<
4 (
5 select avg(salary)
6 from employees
7 where department_id=e.department_id
8 );
```

EMPLOYEE_ID	LAST_NAME	SALARY
101	Yang	17000
102	Garcia	17000
105	Williams	4800
106	Jackson	4800
107	Nguyen	4200
110	Chen	8200
111	Sciarra	7700
112	Urman	7800
113	Popp	6900
115	Khoo	3100
116	Baida	2900

EMPLOYEE_ID	LAST_NAME	SALARY
117	Tobias	2800
118	Hieuro	2600
119	Colmenares	2500
125	Nayer	3200
126	Rikkilineni	2700
127	Landry	2400
128	Markle	2200
129	Bissot	3300
130	Atkinson	2800
131	Harlow	2500
132	Olson	2100

EMPLOYEE_ID	LAST_NAME	SALARY
133	Mallin	3300
134	Rogers	2900
135	Gee	2400
136	Philtanker	2200
138	Stiles	3200
139	Seo	2700
140	Patel	2500
142	Davies	3100
143	Matos	2600
144	Vargas	2500
153	Olsen	8000

EMPLOYEE_ID	LAST_NAME	SALARY
154	Cambault	7500
155	Tuvault	7000
159	Smith	8000
160	Doran	7500
161	Small	7000
164	Marvins	7200
165	Lee	6800
166	Ande	6400

12.

Write a query to display the LOC and average salary of Each location.(Scalar Sub query).

```
SQL> select city,
2  (
3  select avg(salary)
4  from employees
5  where department_id in
6  (
7  select department_id
8  from departments
9  where location_id=l.location_id
10 )
11 ) avg_salary
12 from locations l;

CITY                AVG_SALARY
-----
Seattle              8845.33333
Toronto              9500
London               6500
South San Francisco  3475.55556
Southlake            5700
Munich               10000
Oxford               8955.88235
Sydney
Singapore
Rome
South Brunswick

CITY                AVG_SALARY
-----
Stretford
Geneva
Utrecht
Mexico City
Sao Paulo
Venice
Whitehorse
Hiroshima
Bern
Tokyo
Beijing

CITY                AVG_SALARY
-----
Bombay

23 rows selected.
```

13.

Write a query to display the least N salaries. Use In-Line Views.

```
SQL> select *
2  from
3  (
4  select employee_id,last_name,salary
5  from employees
6  order by salary
7  )
8  where rownum<=5n;
Enter value for n: 4
old 8: where rownum<=5n
new 8: where rownum<=4

EMPLOYEE_ID LAST_NAME          SALARY
-----
132 Olson                2100
128 Markle               2200
136 Philtanker           2200
127 Landry               2400
```

Display the Last N rows from the employees table(Use Correlated Sub Queries).

14.

```

SQL> select *
2  from employees e
3  where &n>
4  (
5  select count(*)
6  from employees
7  where employee_id>e.employee_id
8  )
9  order by employee_id;
Enter value for n: 3
old 3: where &n>
new 3: where 3>

```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY
204	Hermann	Brown					
HBROWN				1.515.555.0169	07-JUN-12	PR_REP	10000
				101 70			
205	Shelley	Higgins					
SHIGGINS				1.515.555.0170	07-JUN-12	AC_MGR	12008
				101 110			
206	William	Gietz					
WGIETZ				1.515.555.0171	07-JUN-12	AC_ACCOUNT	8300
				205 110			

15.

Display the employees from employees table and sort only employees working in DALLAS.(Scalar Sub Query).

```

no rows selected
SQL> select employee_id,last_name,department_id
2  from employees
3  where department_id in
4  (
5  select department_id
6  from departments
7  where location_id=
8  (
9  select location_id
10 from locations
11 where city='Dallas'
12 )
13 );
no rows selected

```

16.

Display the employees who earn a salary less than avgsal of their respective department. Also display the avgsal.(Use Inline Views).

```

SQL> select e.employee_id,
2      e.last_name,
3      e.salary,
4      d.avg_sal
5  from employees e,
6  (
7  select department_id,
8  avg(salary) avg_sal
9  from employees
10 group by department_id
11 ) d
12 where e.department_id=d.department_id
13 and e.salary<d.avg_sal;

```

EMPLOYEE_ID	LAST_NAME	SALARY	AVG_SAL
101	Yang	17000	19333.3333
102	Garcia	17000	19333.3333
105	Williams	4800	5760
106	Jackson	4800	5760
107	Nguyen	4200	5760
110	Chen	8200	8601.3333
111	Sciarra	7700	8601.3333
112	Urman	7800	8601.3333
113	Popp	6900	8601.3333
115	Whoo	3100	4150
116	Baida	2900	4150
117	Tobias	2800	4150
118	Himuro	2600	4150
119	Colmenares	2500	4150
125	Nayer	3200	3475.5556
126	Mikkilineni	2700	3475.5556
127	Landry	2400	3475.5556
128	Markle	2200	3475.5556
129	Bissot	3300	3475.5556
130	Atkinson	2800	3475.5556
131	Marlow	2500	3475.5556
132	Olson	2100	3475.5556
133	Mallin	3300	3475.5556
134	Rogers	2900	3475.5556
135	Gee	2400	3475.5556
136	Philtanker	2200	3475.5556
138	Stiles	3200	3475.5556
139	Seo	2700	3475.5556
140	Patel	2500	3475.5556
142	Davies	3100	3475.5556
143	Matos	2600	3475.5556
144	Vargas	2500	3475.5556
153	Olsen	8000	8955.88235

17.

Write a query that display the LOC of those DEPTS that have a sum of sal less than the avgsal of all the employees in DEPT table. (Use WITH Clause).

```

SQL> with dept_sal as
2  (
3  select department_id,
4  sum(salary) total_sal
5  from employees
6  group by department_id
7  )
8  select city
9  from locations
10 where location_id in
11 (
12 select location_id
13 from departments
14 where department_id in
15 (
16 select department_id
17 from dept_sal
18 where total_sal<
19 (
20 select avg(salary)
21 from employees
22 )
23 )
24 );

```

CITY

Seattle